OD OpsDesk for TBO

TRACK 1 · ZENDESK REPLACEMENT FOR TRAVEL OPS

OpsDesk for TBO

A booking-centric operations platform: the booking is the context, the ticket is one interaction.

Working product · synthetic demo dataset · figures measured on this build

- Six connected workspaces — queue, case, routing, escalation, intelligence, simulation.
- Eight canonical intents classified by a calibrated in-house model.
- SLA that respects cancellation deadlines and check-in dates.

THE THESIS

A booking is the operational context

A ticket is only one interaction associated with that booking.

TICKET-CENTRIC TODAY

- Every refund, voucher or amendment arrives as an isolated ticket.
- The agent rebuilds the booking context by hand, every time.
- Supplier conversations leak between groups; SLA ignores travel dates.
- Nothing connects recurring supplier failures to the product team.

BOOKING-FIRST WITH OPSDESK

- The TBO booking id is extracted from the message and becomes the case anchor.
- Every related ticket, supplier thread and SLA deadline lives on one case graph.
- Routing, escalation and analytics all operate on that same graph.
- Recurring patterns become product bug proposals with evidence.

One data model, one audit trail: the booking id is the join key across tickets, conversations, routing audits and exports.

THE PRODUCT

Six workspaces, one operating model

Each workspace is a real screen backed by the same API — no static mock-ups.

W1

Ops Queue & SLA

Live countdowns, governing rule per ticket, regional holiday handling.

W2

Case Workspace

Three-pane case: thread, booking context, independent side conversations.

W3

Routing Studio

Score breakdown out of 100, exclusions, immutable audit.

W4

War Room

Escalated and at-risk inventory with real ownership handover.

W5

Intelligence Hub

Themes, bug proposals, streamed governance exports.

W6

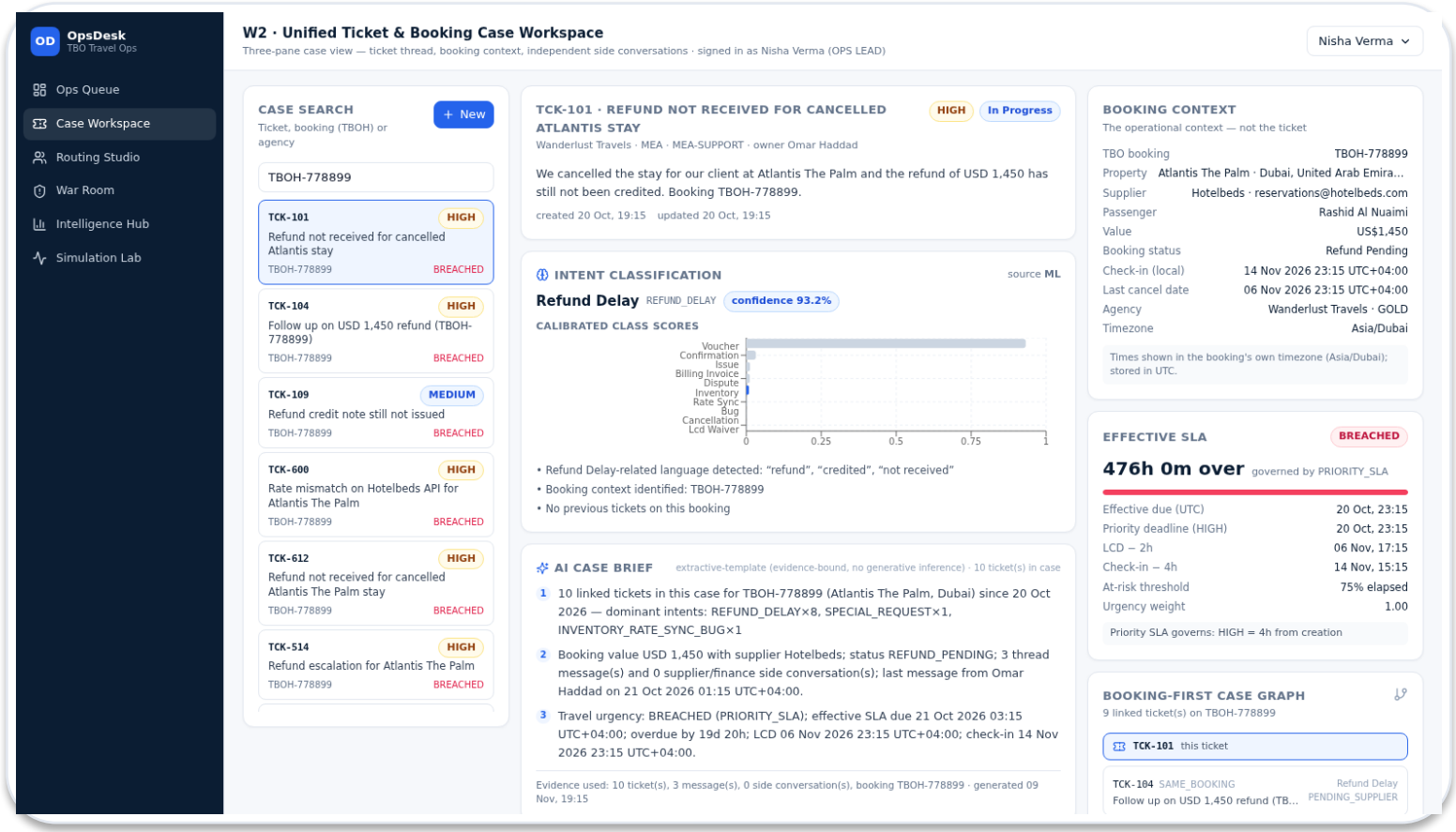
Simulation Lab

Five seeded journeys executed against the live API.

WORKSPACE 2

The case workspace

Booking context, classification, evidence-bound brief and thread in one view.



Live screen — TCK-101 on TBOH-778899 (Wanderlust Travels, MEA).

- Stage A regex extracts TBOH-###### from free text when the field is empty.
- Stage B assigns one of eight intents with a calibrated confidence.
- Related tickets on the same booking form the case graph.
- The brief has exactly three points and cites the tickets it used.

INTELLIGENCE

Understanding the intent

TF-IDF + logistic regression, calibrated, with a rule-based fallback.

HOLDOUT ACCURACY

0.8958

template-disjoint · n=288

MACRO F1

0.8964

8 intents · 576 train rows

TRAINING DATA

Synthetic

no real TBO ticket text used

FALLBACK

Rules

pipeline survives a missing model

Intent	Precision	Recall	F1	Support
REFUND_DELAY	0.9677	0.8333	0.8955	36
CANCELLATION_LCD_WAIVER	0.9231	1.0000	0.9600	36
VOUCHER_CONFIRMATION_ISSUE	0.8889	0.8889	0.8889	36
AMENDMENT_NAME_DATE	0.9211	0.9722	0.9459	36
INVENTORY_RATE_SYNC_BUG	1.0000	0.8333	0.9091	36
SPECIAL_REQUEST	0.7805	0.8889	0.8312	36
SUPPLIER_OVERBOOKING_RELOCATION	0.8000	0.8889	0.8421	36
BILLING_INVOICE_DISPUTE	0.9394	0.8611	0.8986	36

WHY IT MATTERS

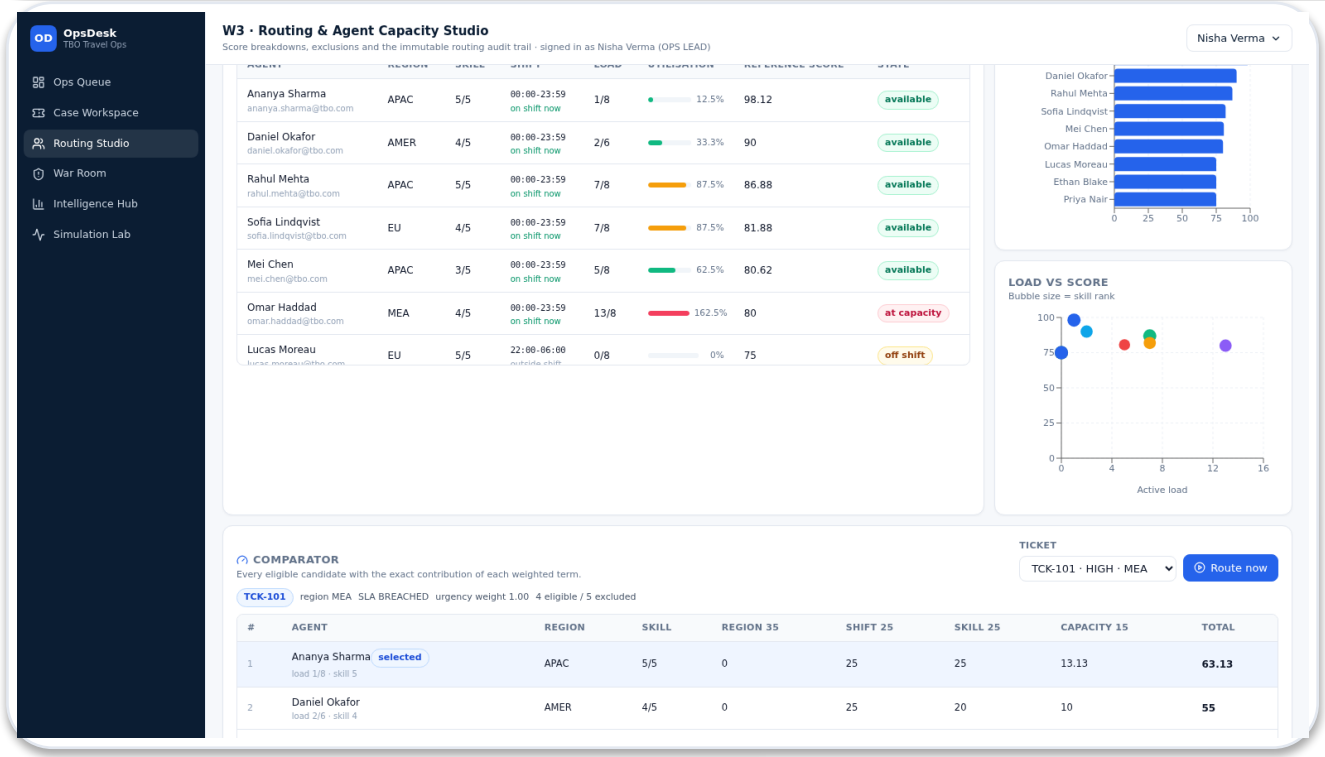
- Intents drive routing, SLA urgency and analytics — not free-text tags.
- Confidence is the calibrated model output, never a constant.
- Agents can override, and the override is audited.

WORKSPACE 3

Routing you can audit

Hard filters first, then a 100-point score with every term stored.

```
35*RegionMatch + 25*ShiftMatch + 25*(SkillRank/5)*UrgencyWeight + 15*(1 - ActiveLoad/MaxCapacity)
```



Live screen — agent board, comparator and audit trail.

- Exclusions are explicit: offline, at capacity, off shift, skill floor, regional eligibility.
- Tie-break: highest score → lowest active load → lowest agent id.
- Comparator median latency measured at 3.92 ms (p95 4.49 ms).
- Each run writes an append-only audit row; overrides are separate audited rows.

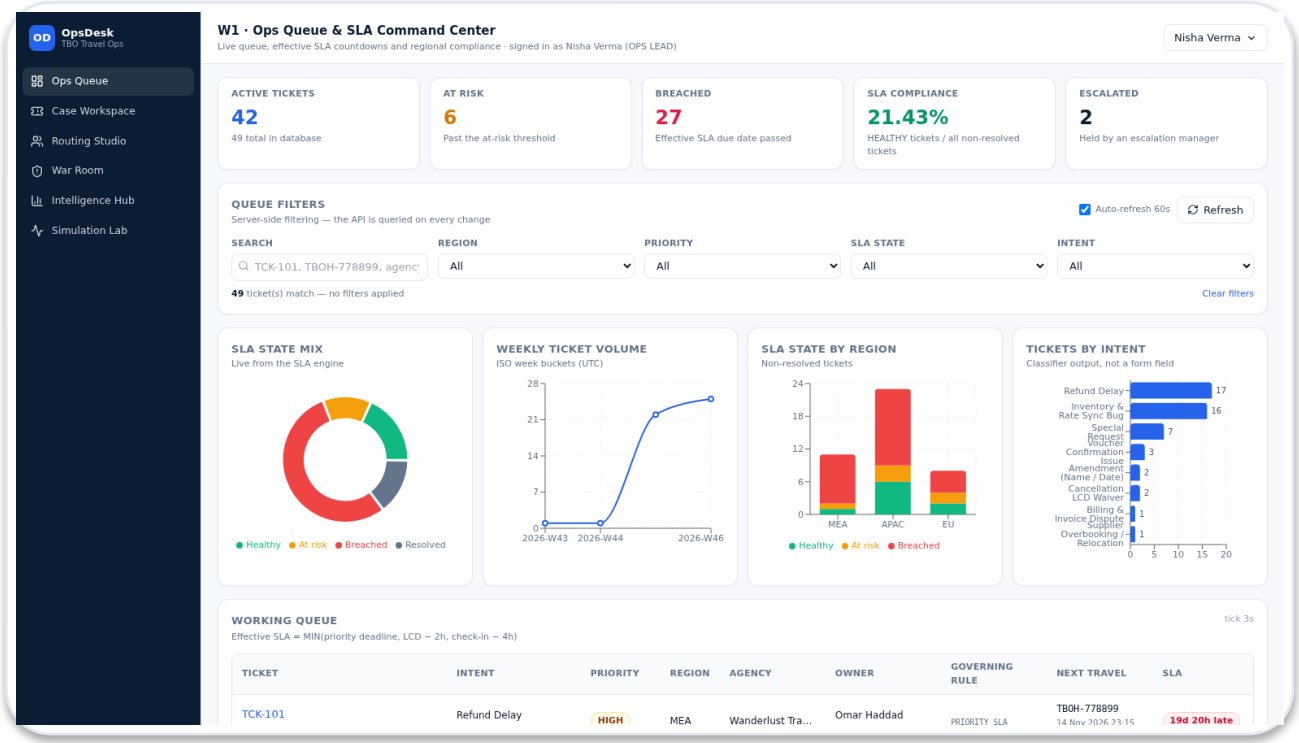
WORKSPACE 1 & 4

SLA that respects the travel

The clock is the earliest of the priority SLA and the travel deadlines.

EffectiveSLADueAt =
MIN(PriorityDeadline, LCD − 2h, CheckIn − 4h)

Priority	SLA window	At-risk threshold
CRITICAL	1 hour	75% elapsed
HIGH	4 hours	75% elapsed
MEDIUM	12 hours	75% elapsed
LOW	24 hours	75% elapsed
Regional holiday	unchanged	50% elapsed (recorded)



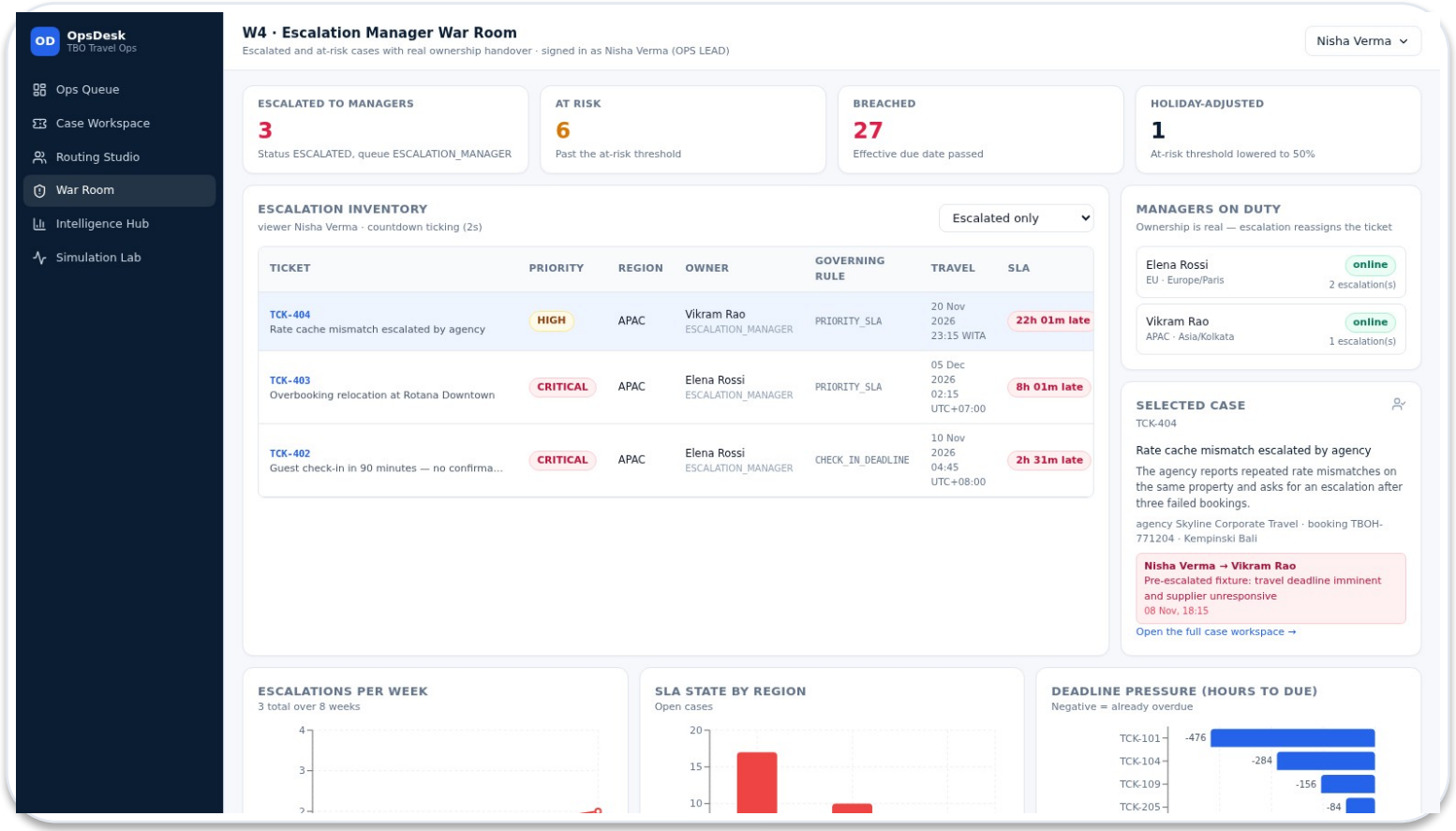
Live screen — queue with per-ticket governing rule and countdown.

Missing or already-expired travel dates are reported explicitly and never zeroed. Times are stored in UTC and displayed in the booking's timezone.

WORKSPACE 4

Escalation with real ownership

Escalate to a manager: reason, eligible manager, new queue, immutable event.



Live screen — escalation inventory, managers on duty, nearest deadlines.

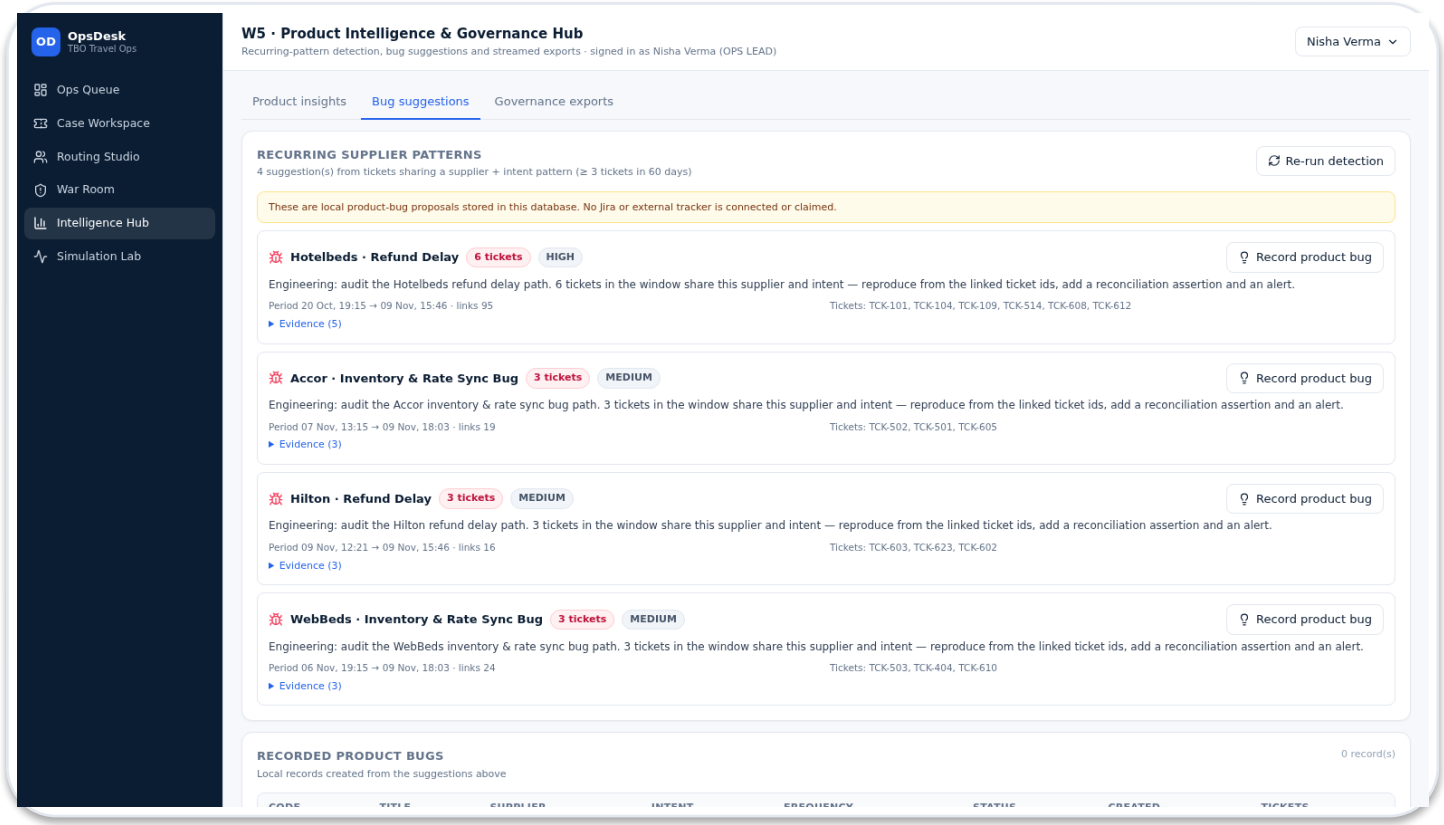
WHAT HAPPENS

- Status becomes ESCALATED and the queue becomes ESCALATION_MANAGER.
- Ownership moves to the eligible manager in one atomic transaction.
- escalated_by / escalated_at and the reason are stored.
- The ticket timeline records the event permanently.

WORKSPACE 5

From operations to product

Recurring supplier + intent patterns become bug proposals with evidence.



Live screen — suggestions computed from tickets sharing supplier + intent.

DETECTION RULE

Group tickets by (supplier, intent), require ≥ 3 in the window, attach the ticket ids, period, frequency and evidence, then propose an action.

SCOPE DISCIPLINE

Records are created in this database only. No Jira or external tracker integration exists or is claimed.

GOVERNANCE

50,000 rows, streamed and verified

A server-side iterator writes real bytes; the file is validated by re-reading it.

ROWS IN FILE

50,000

25 columns

SERVER DURATION

1,289.12 ms

38,786 rows/s

FILE SIZE

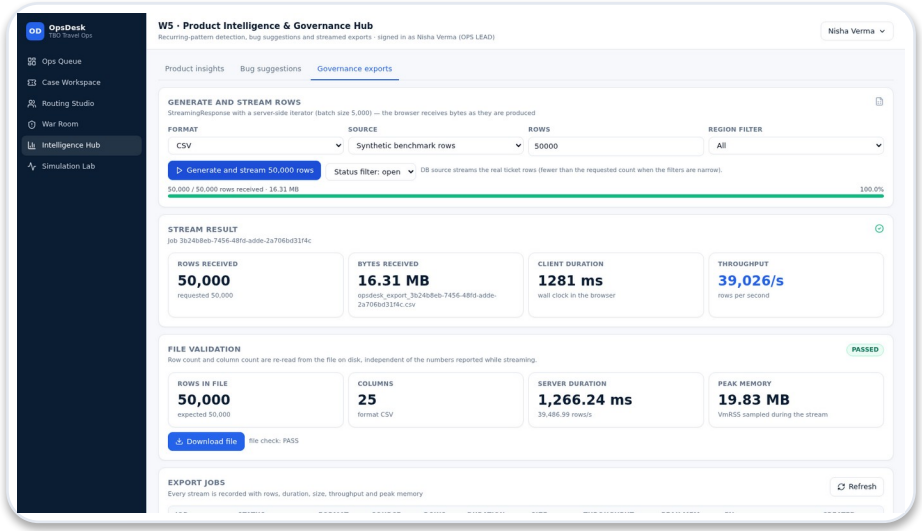
16.313 MB

CSV, streamed

PEAK MEMORY

8.848 MB

VmRSS sampled



Live screen — progress, measured metrics and file validation.

- StreamingResponse over an iterator, batches of 5,000.
- Progress reflects bytes actually received.
- Validation re-reads the file: row count, columns, readability.

EVIDENCE

Measured, not asserted

Every figure on this slide comes from an artefact produced by this build.

Check	Result	Source
Automated test suite	106 passed	pytest tests backend/tests
Seeded journeys	5/5 passed · 34/34 steps	simulation/run_all
Classifier holdout	0.8958 accuracy · 0.8964 macro F1	ai_models/evaluation/metrics.json
Export integrity	50,000 rows · 25 cols · PASS	benchmark_report.json
Browser crawl	6 workspaces · 0 console errors	scripts/ui_crawl.py
Routing comparator	3.92 ms median	benchmark_report.json

Not claimed: real TBO production data, live supplier APIs, or any external issue-tracker integration. The dataset is synthetic and labelled as such throughout the product.

OpsDesk does not just manage tickets. It understands the entire travel operation.

— Booking-first data model with an auditable case graph

— Calibrated intent classification and transparent 100-point routing

— Travel-aware SLA, transfer-safe conversations, measured governance exports